

STA 610 Data analysis report - Sparrow

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Abstract

This report designs an experimental framework to examine the effects of age, competition from other females, and other factors on each bird's number of offspring. The structure is organized as follows. In section 1, the objective of this experimental study is outlined, focusing on identifying potential factors contributing to variability in the number of offspring. section 2 discusses the study design, sample size, and the approach to determining potential micro- and macro-level effects. In section 3, the quality of the data is evaluated, and ad hoc strategies for data cleaning and wrangling are proposed to address missing values and anomalies present in the dataset. Specifically, subsection 3.3 examines the distribution of **spf** (number of offspring) and its properties after transformation, which are subsequently used to test for normality assumptions with qqplots of residuals. Additionally, the section analyzes potential grouping effects while accounting for the collinearity among **cohort**, **year**, and **age**. section 4 focuses on the model fitting process, starting with a basic linear model (**lm**), followed by linear mixed-effects models (**lmer**) and generalized linear mixed-effects models (**glmer**). While increasing model complexity may reduce the Bayesian Information Criterion (BIC), the paper emphasizes the importance of incorporating grouping effects and evaluates the statistical significance of adding random effects or random slopes based on hypothesis testing. Finally, section 5 and section 6 summarize the findings and propose potential improvements for further analysis.

1 Introduction

This report uses the sparrow dataset, which captures the pattern of offspring on an island in the Pacific Northwest over 19 years. However, there are discrepancies with the information provided in Dr. Peter Hoff's statement, as the dataset used in this report contains 18 years of observations, with year 6 missing. The goal is to extrapolate the relationships among factors contributing to the variability in the number of offspring. The paper adopts the Hierarchical Model methodology outlined by [Hoff, 2022], [Chatterjee and Hadi, 2008], and the hypothesis testing strategy addressed by [Seber, 2015]. The core objective is not only to apply the basic framework of the linear model but also to test for normality, heteroscedasticity, and macro-level assumptions. Ignoring grouping effects could lead to several problems, such as violations of the independence assumption, biased estimators, and misestimated residual variance. Therefore, the optimal approach is to identify a model that provides the best estimators while ensuring that residuals do not exhibit any statistically observable patterns.

2 Study Design

This section discussed the methodology used to investigate the variability in the number of offspring (**spf**) and the effects of macro- and micro-level explanatory variables. The process emphasizes a step-by-step approach to data cleaning, exploratory analysis, and hierarchical modeling.

2.1 Data Cleaning and Transformation

The analysis begins with data cleaning to address missing values, inconsistencies, and anomalies. Year 6 is entirely missing from the dataset, and other irregularities are resolved as part of this step. Spatial coordinates (**x** and **y**) are used to generate grouping variables, dividing the study area into distinct zones

(e.g., Area 1, 2, 3, 4) to account for spatial heterogeneity. This transformation ensures that location-based effects are captured in the subsequent modeling process.

2.2 Preliminary Analysis of Variables

The properties of both the response variable (**spf**) and explanatory variables are examined in preparation for modeling. Macro-level variables such as **fpop**, **x**, **y**, and **year** are analyzed to identify population and environmental influences. Micro-level variables, including **age**, **cohort**, and **band**, are studied to capture individual-level effects. Initial analysis highlights the collinearity between **age**, **cohort**, and **year**, necessitating the careful selection of variables for modeling.

2.3 Hierarchical Modeling Approach

The modeling process is designed to test assumptions systematically and account for grouping effects:

- **Step 1: Linear Model (lm):** A linear model is fitted as the baseline. Residuals are assessed using QQ plots to test the normality assumption. Patterns in the residuals may suggest violations of normality and highlight potential grouping effects, such as dependencies within groups defined by **band** or **cohort**.
- **Step 2: Linear Mixed-Effects Model (lmer):** To address grouping effects and dependency, a linear mixed-effects model is fitted, incorporating random intercepts and slopes. While **lmer** accounts for group-level variability, it still relies on the normality assumption. Likelihood ratio tests (LRTs) are used to compare nested models, starting with a simple structure and adding random intercepts and slopes incrementally.
- **Step 3: Generalized Linear Mixed-Effects Model (glmer):** If the normality assumption is significantly violated, a generalized linear mixed-effects model is employed. This approach allows for alternative response distributions (e.g., Poisson or Negative Binomial), making it suitable for count data like **spf**. LRTs are again used to evaluate the significance of adding random effects and interactions.

2.4 Macro- and Micro-Level Effects

Macro-level variables (**fpop**, **x**, **y**, **year**) capture environmental and population-level influences, while micro-level variables (**age**, **cohort**, **band**) represent individual reproductive characteristics. By modeling these effects hierarchically, the study aims to quantify their contributions to the variability in **spf** while addressing assumptions of normality, independence, and grouping effects. This design provides a structured approach to understanding the variability in **spf**, balancing the exploration of individual- and population-level factors with the systematic testing of model assumptions.

3 Sparrow Dataset Description

3.1 Missing Value

- **Problem:** One of the main challenges with this dataset is the presence of missing values, particularly in the first year of the study. When the research was initially conducted on an island in the Pacific Northwest, the researchers did not know the ages of the birds that were already present. Consequently, even though the relationship $\text{cohort} = \text{year} - \text{age} + 1$ is known, imputing the missing values for the first year is problematic due to a lack of reliable information. However, potential approaches include estimating the missing ages based on the number of offspring.
- **Solution:** To examine the effect of **age** and other factors on the number of offspring, this paper aims to determine whether these variables have a positive or negative impact and to quantify their

effects. Missing data for **cohort** and **age** occur simultaneously, primarily in the first year of the study, where observations are also less reliable due to initial unpreparedness. Given these challenges, this paper proposes dropping all missing values from the first year. Additionally, variables such as **x**, **y**, and **band** have 1, 5, and 5 missing values, respectively, which are too few to justify meaningful imputation. Therefore, the final strategy adopted is to remove all rows with missing values, ensuring the analysis focuses on more complete and reliable data to better understand the factors influencing the number of offspring.

Description	Count
Number of rows with exactly one NA (cohort, year, age)	0
Number of rows with exactly two NAs (cohort, year, age)	101
Number of rows with exactly three NAs (cohort, year, age)	0
Number of rows with exactly one NA (whole dataset)	0
Number of rows with exactly two NAs (whole dataset)	105
Number of rows with exactly three NAs (whole dataset)	1
Number of rows containing at least one NA (whole dataset)	106

Table 1: Summary of Missing Rows

band	cohort	year	fpop	age	spf	x	y
1	101	0	0	101	0	5	5

Table 2: Number of NAs for Each Variable

Year	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
Count	34	26	21	7	4	0	2	0	0	1	0	1	1	0	0	3	0	3	3

Table 3: Number of Rows with at Least One NA for Each Year

3.2 Unreasonable Structure

- **Problem 1:**

For the band **54544**, an anomaly is observed in the data. The bird was observed in **year = 16** with **fpop = 10** and **spf = 9**, but both **cohort** and **age** were missing. Despite being observed in that year, it was not tagged, which is a clear inconsistency. Instead, the bird was tagged in the following year (**year = 17**) as **cohort = 17** with **age = 1**, which is incorrect because it implies that the bird was a new individual. This tagging issue highlights a discrepancy in the data collection process and suggests that records for **band 54544** in the first observation year were either incomplete or mishandled.

Band	Cohort	Year	Fpop	Age	Spf	X	Y
54544	NA	16	10	NA	9	25.7500	3.9167
54544	17	17	28	1	8	24.9000	4.3833
54544	NA	18	42	NA	5	24.8750	4.2750
54544	NA	19	42	NA	4	28.3125	3.9375

Table 4: Anomalous Records for Band 54544

- **Problem 2:**

For the band **54533**, an inconsistency is observed in the **cohort**, which represents the year in which the bird was tagged. While the **cohort** is consistently recorded as **16** in the first two rows, it is incorrectly recorded as **15** in the third row, corresponding to **year = 18**. This inconsistency suggests an error in the data recording process, as the **cohort** for the same bird should remain constant across all observations. Such discrepancies require careful review to ensure the accuracy of the dataset.

Band	Cohort	Year	Fpop	Age	Spf	X	Y
54533	16	16	10	1	8	21.7500	2.8125
54533	16	17	28	2	5	21.6500	3.7167
54533	15	18	42	4	6	21.3500	3.5750

Table 5: Anomalous Records for Band 54533

- **Solution** For scenarios like Problem 1, the issue is automatically resolved by dropping the rows with missing values. As a result, the conflict in chronological order no longer exists in the dataset. For Problem 2, since this is the only case, I manually corrected the **Cohort** to 16 and updated the **Year** to 19 to maintain consistency. However, it is recommended to consult the data collector regarding this anomaly. The focus remains on obtaining clean data and minimizing the likelihood of outliers and irregular structures, ensuring the integrity of the analysis.

3.3 Exploratory Data Analysis(EDA)

3.3.1 spf: number of offspring (“sparrows fledged”)

The original response variable, **spf**, is count data with a right-skewed distribution and potential outliers, shown in Fig. 1. Both log and square root transformations improve symmetry but do not fully meet normality assumptions, shown in Figs. 2–3. Since **spf** is non-negative, it is suitable for Poisson or Negative Binomial distributions in a generalized linear mixed-effects model (GLMM). Preliminary analysis suggests overdispersion may be a concern, making Negative Binomial more appropriate. For simpler analyses, linear models with transformations can approximate normality. Hierarchical structures in the data will likely require GLMMs with a log-link to properly model variance while accounting for the nesting of observations.

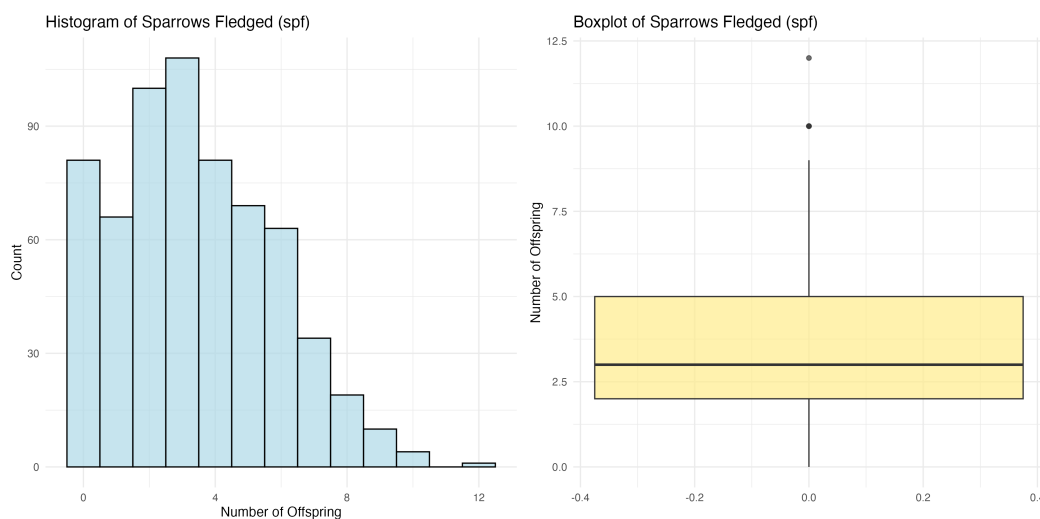


Figure 1: Distribution and summary of the number of offspring (**spf**)

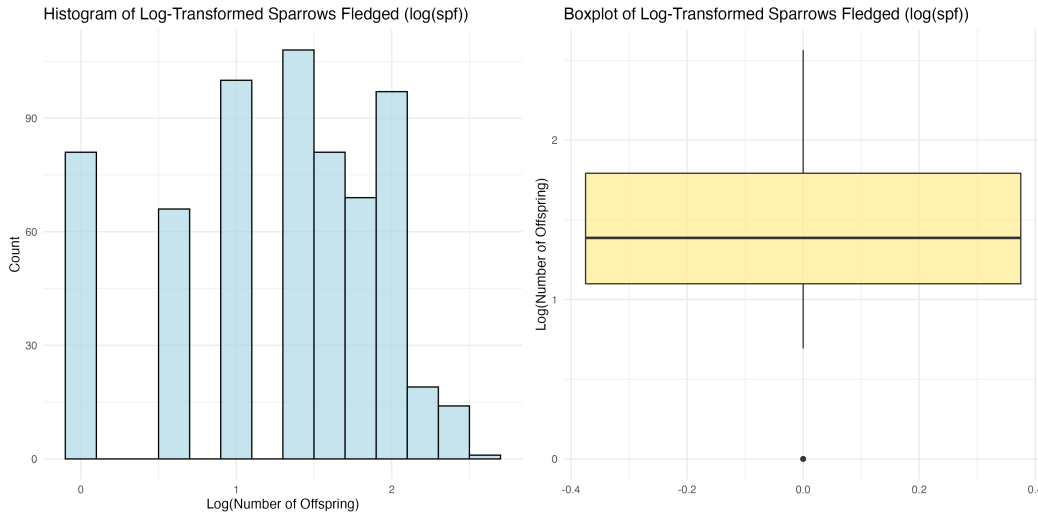


Figure 2: Distribution and summary of the log-transformed number of offspring ($\log(\text{spf})$)

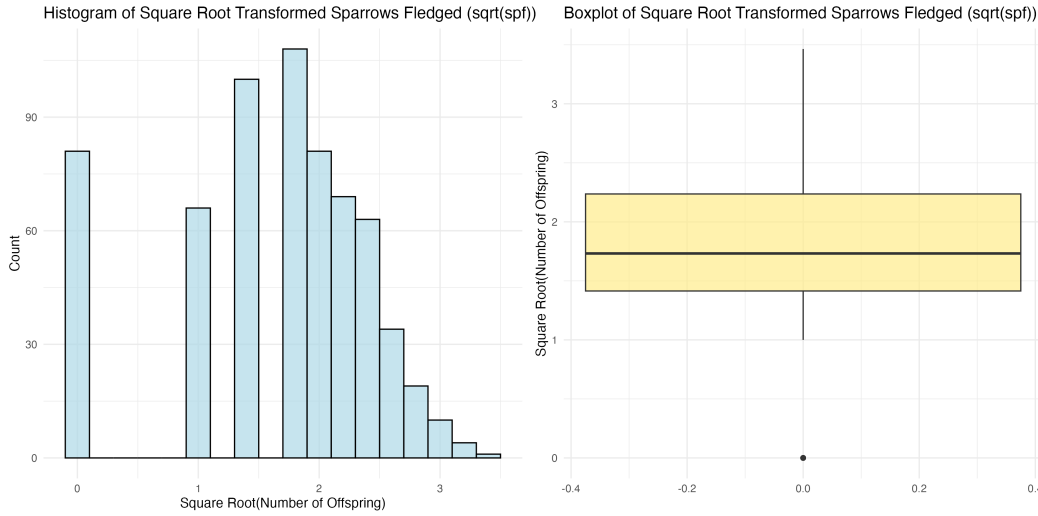


Figure 3: Distribution and summary of the square root-transformed number of offspring ($\sqrt{\text{spf}}$)

3.3.2 Relationship Between spf and Year, Cohort, and age

The boxplots illustrate the variability in the number of offspring (spf) across **age**, **cohort**, and **year**, providing insights into potential grouping effects. Each variable shows distinct patterns of variability, which may inform hierarchical modeling decisions. Although **cohort**, **year**, and **age** are collinear (any one can be determined from the other two), this analysis focuses on understanding their individual grouping effects before deciding which variable to include in the final model.

- **Age:** Younger sparrows (ages 1 and 2) tend to have higher variability in spf , with median values decreasing slightly as age increases. Sparrows older than 5 show a sharp decline in offspring numbers, with limited observations for these older birds.
- **Cohort:** The distribution of spf varies significantly across cohorts, with some cohorts consistently showing higher median values and others showing lower medians or increased variability. This suggests that cohort effects might play a role in influencing reproductive success.
- **Year:** The distribution of spf appears uneven across years, with some years exhibiting greater variability and others showing narrower ranges. This temporal variability may capture changes in environmental or population conditions influencing offspring numbers.

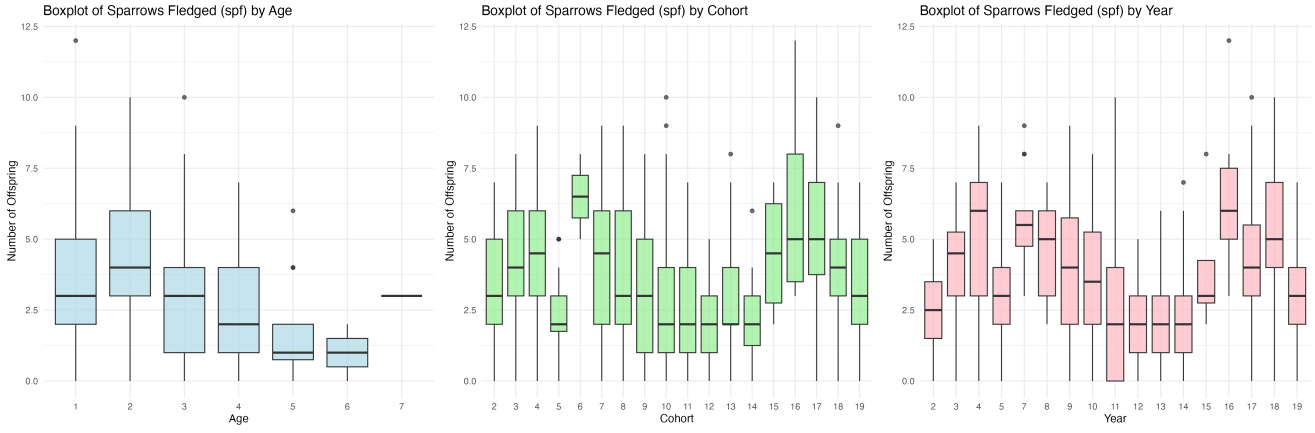


Figure 4: Boxplots of **spf** by **age** (left), **cohort** (center), and **year** (right), showing variation across grouping factors.

3.3.3 Relationship Between spf and fpop, x, and y

The scatterplots illustrate the relationships between the number of offspring (*spf*) and three key explanatory variables: the female population (*fpop*), the x-coordinate (*x*), and the y-coordinate (*y*). A linear regression line (*geom_smooth(method = "lm")*) is overlaid on each plot to highlight trends.

- *Female Population (fpop)*: A noticeable negative relationship exists between *spf* and *fpop*. The linear regression line indicates that as the number of females in the population increases, the number of offspring per sparrow decreases. This trend likely reflects the impact of competition for resources, mates, or nesting sites, which becomes more pronounced in larger populations.
- *X-Coordinate (x)*: The x-coordinate shows a weak, slightly negative trend with *spf*, as indicated by the linear regression line. This subtle pattern could suggest spatial variability in resource availability or environmental quality across the island's horizontal axis, although the effect appears minor.
- *Y-Coordinate (y)*: The y-coordinate displays an almost flat relationship with *spf*, as reflected by the near-horizontal regression line. This suggests that the vertical spatial distribution has little to no systematic influence on reproductive success, implying uniformity in environmental conditions along the island's vertical axis.

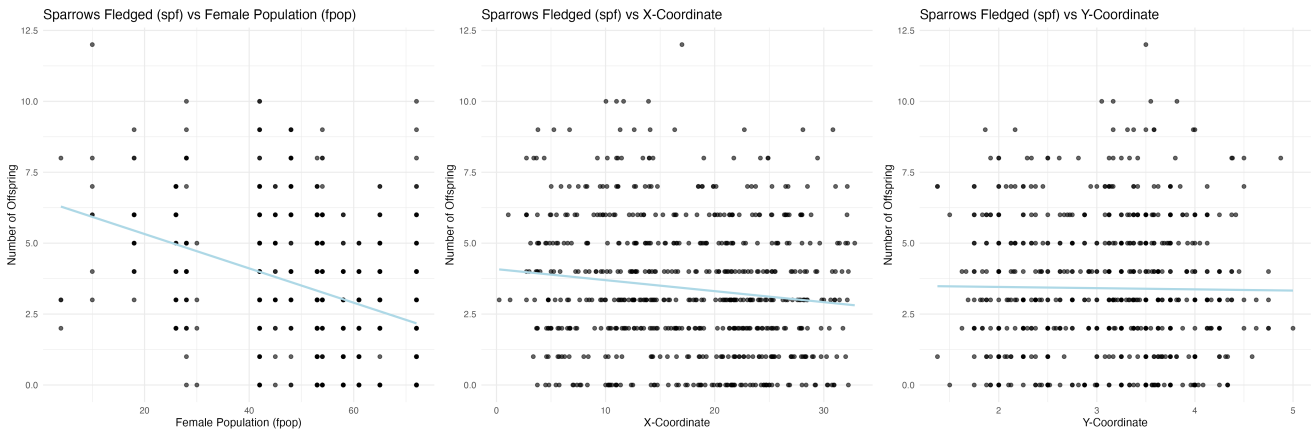


Figure 5: Scatterplots of *spf* against *fpop*, *x*, and *y*, with linear regression lines (*geom_smooth(method = "lm")*) indicating trends.

3.4 Collinearity

The correlation heatmap shows the core relationships among the numeric variables. A strong collinearity exists between **cohort** and **year** ($r = 0.97$), which is expected due to their linear dependency by definition. Interestingly, **age**, despite being derived from **cohort** and **year** ($\text{cohort} = \text{year} - \text{age} + 1$), does not exhibit strong collinearity with these variables ($r = 0.21$ with **year** and $r = -0.08$ with **cohort**). This weaker relationship can be attributed to the variability in ages observed across different years and cohorts, which breaks the direct proportionality. Additionally, **spf** (number of offspring) shows a moderate negative correlation with **fpop** (number of females in the population), likely due to competitive pressures, while spatial coordinates (**x**, **y**) exhibit minimal correlations with other variables, suggesting a limited direct influence on offspring variability. These observations are crucial for selecting variables for modeling while addressing multicollinearity concerns, especially in hierarchical or mixed-effects frameworks.

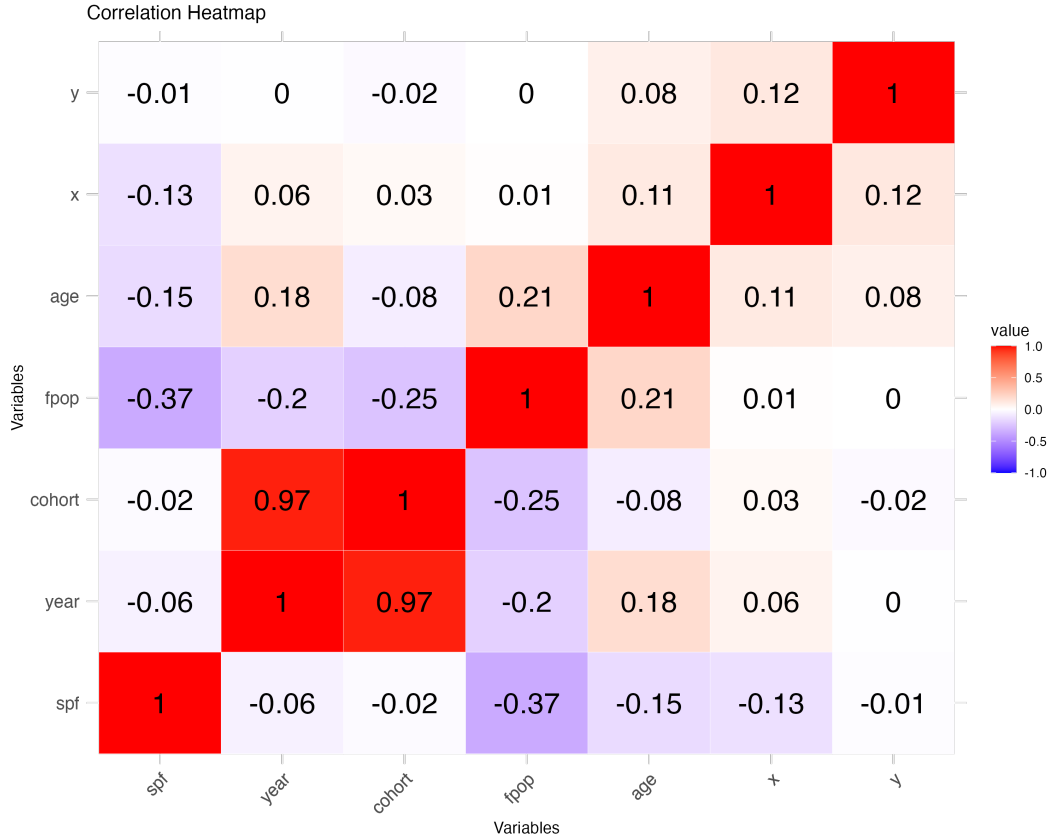


Figure 6: Heatmap of correlation coefficients among numeric variables, illustrating strong collinearity between **cohort** and **year**, as well as other notable relationships. The color gradient highlights positive correlations in red and negative correlations in blue.

3.5 Heterogeneity Across Groups

- **Methodology:** To assess the homogeneity of variances across groups, the paper didn't use the package to do the testing. Instead, this paper follow the methodology from [Hoff, 2022], the Levene's test is implemented as follows:

1. **Model the Response Variable:** Fit a linear model for the response variable $y_{i,j}$, where j represents the group:

$$y_{i,j} = \mu_j + \epsilon_{i,j}, \quad \epsilon_{i,j} \sim N(0, \sigma^2)$$

Here, μ_j is the mean for group j , and $\epsilon_{i,j}$ are residuals.

2. **Compute Absolute Residuals:** Calculate the absolute residuals for each observation:

$$z_{i,j} = |y_{i,j} - \hat{\mu}_j|$$

where $\hat{\mu}_j$ is the estimated group mean.

3. **Model the Absolute Residuals:** Fit another linear model using the absolute residuals $z_{i,j}$ as the response:

$$z_{i,j} = \nu_j + \eta_{i,j}$$

where $\nu_j = \mathbb{E}[z_{i,j} \mid G = j]$ and $\eta_{i,j}$ are the residuals.

4. **ANOVA F-Test:** Test the null hypothesis:

$$H_0 : \nu_1 = \nu_2 = \dots = \nu_k$$

where k is the number of groups. Under H_0 , the variances across groups are equal.

5. **F-Statistic:** Calculate the F-statistic as:

$$F = \frac{\text{Between-group variation}}{\text{Within-group variation}}$$

Compare F to the critical value from the F-distribution to determine if H_0 is rejected.

- **Testing Result:**

Grouping Variable	Df	Sum Sq	Mean Sq	F value (Pr(>F))
Age	6	25.23	4.2043	2.5231 (0.0202) *
Cohort	17	51.07	3.0043	2.0582 (0.0074) **
Year	16	65.67	4.1043	2.9001 (0.0001) ***
Residuals (Age)	629	1048.09	1.6663	-
Residuals (Cohort)	618	902.08	1.4597	-
Residuals (Year)	619	876.03	1.4152	-

Note: Significance codes: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

Table 6: Levene's Test Results for Variance Homogeneity Across Age, Cohort, and Year

Hence, under this testing result, it is more likely for the heteroscedasticity across groups to stem from **Year** variable. Which order the grouping factor based on the statistical significance for the following model fitting and diagnostic process.

4 Model fitting and Diagnostics

4.1 Linear Model

The following are the mathematical formulations for the four linear models analyzed in this study:

- **Model 1: Full Linear Model (fit1m0)**

$$\text{spf}_i = \beta_0 + \beta_1 \cdot \text{year}_i + \beta_2 \cdot \text{fpop}_i + \beta_3 \cdot x_i + \beta_4 \cdot y_i + \epsilon_i, \quad \epsilon_i \sim N(0, \sigma^2)$$

- **Model 2: Reduced Linear Model Without y (fit1m1)**

$$\text{spf}_i = \beta_0 + \beta_1 \cdot \text{year}_i + \beta_2 \cdot \text{fpop}_i + \beta_3 \cdot x_i + \epsilon_i, \quad \epsilon_i \sim N(0, \sigma^2)$$

- **Model 3: Linear Model With Log Transformation of $\text{spf} + 1$ (fitlm2)**

$$\log(\text{spf}_i + 1) = \beta_0 + \beta_1 \cdot \text{year}_i + \beta_2 \cdot \text{fpop}_i + \beta_3 \cdot x_i + \epsilon_i, \quad \epsilon_i \sim N(0, \sigma^2)$$

- **Model 4: Linear Model With Square Root Transformation of $\text{spf} + 1$ (fitlm3)**

$$\sqrt{\text{spf}_i + 1} = \beta_0 + \beta_1 \cdot \text{year}_i + \beta_2 \cdot \text{fpop}_i + \beta_3 \cdot x_i + \epsilon_i, \quad \epsilon_i \sim N(0, \sigma^2)$$

Call:

```
lm(formula = spf ~ year + fpop + x + y, data = data_drop)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-5.6463	-1.5649	-0.2245	1.4370	7.6883

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	8.090341	0.562407	14.385	< 2e-16 ***
year	-0.066618	0.019288	-3.454	0.00059 ***
fpop	-0.064577	0.006116	-10.558	< 2e-16 ***
x	-0.035325	0.010946	-3.227	0.00132 **
y	-0.002430	0.113538	-0.021	0.98293

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.17 on 631 degrees of freedom

Multiple R-squared: 0.1666, Adjusted R-squared: 0.1613

F-statistic: 31.54 on 4 and 631 DF, p-value: < 2.2e-16

Single term deletions

Model:

	Df	Sum of Sq	RSS	AIC	Pr(>Chi)
<none>			2971.8	990.53	
year	1	56.18	3028.0	1000.44	0.0005579 ***
fpop	1	525.02	3496.8	1092.00	< 2.2e-16 ***
x	1	49.05	3020.8	998.94	0.0012526 **
y	1	0.00	2971.8	988.53	0.9828547

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Table 7: Summary of Linear Model and Variable Selection with **drop1**

The **drop1** function results shown in Table 7 indicate that removing the variable y does not significantly impact the model, as evidenced by its p-value ($p = 0.98293$) and a lower AIC (988.53) compared to the full model. Additionally, due to the observed collinearity among year, cohort, and age, the variable year is selected for inclusion in the model. This choice ensures a more interpretable model while addressing multicollinearity issues.

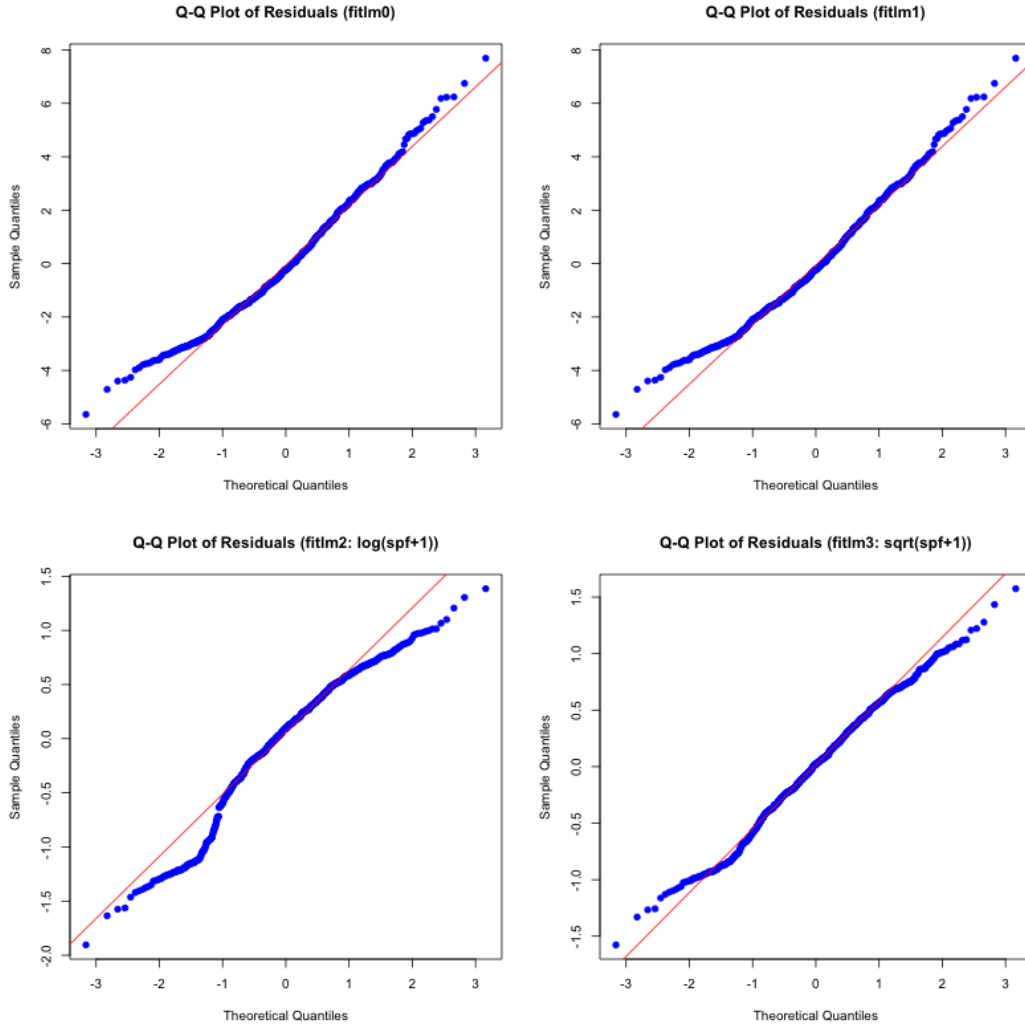


Figure 7: Q-Q plots of residuals for four models: fitlm0 (full model), fitlm1 (reduced model without y), fitlm2 (log-transformed $\text{spf} + 1$), and fitlm3 (square root-transformed $\text{spf} + 1$).

Furthermore, from Fig. 7, it is evident that even after applying transformations to **spf**, the normality assumptions for residuals in the linear model are still not fully satisfied. This issue serves as a key motivation to delve deeper and adopt the **lmer** model, which can account for the grouping effect and provide more reliable inference. Ignoring the grouping effect violates the independence assumption, as sparrows within the same group are expected to exhibit some level of dependency. Addressing this dependency is crucial for drawing valid conclusions.

4.2 Linear Mixed-Effect Model

4.2.1 Random Intercept

The fitted models are as follows:

- Model fitlmer0 (Null Model):

$$\text{spf}_{ij} = \beta_0 + u_j + \epsilon_{ij}$$

where $u_j \sim \mathcal{N}(0, \Psi_u^2)$ represents the random intercept for each year j , and $\epsilon_{ij} \sim \mathcal{N}(0, \sigma^2)$ represents the residual error.

- Model fitlmer1:

$$\text{spf}_{ij} = \beta_0 + \beta_1 \text{fpop}_{ij} + u_j + \epsilon_{ij}$$

where u_j and ϵ_{ij} are defined as in the null model, and β_1 captures the fixed effect of the female population (fpop).

- **Model fitlmer2:**

$$\text{spf}_{ij} = \beta_0 + \beta_1 \text{fpop}_{ij} + \beta_2 x_{ij} + u_j + \epsilon_{ij}$$

where β_2 captures the fixed effect of the x -coordinate (x_{ij}).

- **Model fitlmer3:**

$$\text{spf}_{ij} = \beta_0 + \beta_1 \text{fpop}_{ij} + \beta_2 x_{ij} + \beta_3 y_{ij} + u_j + \epsilon_{ij}$$

where β_3 captures the fixed effect of the y -coordinate (y_{ij}).

- **Model fitlmer4 (Full Model):**

$$\text{spf}_{ij} = \beta_0 + \beta_1 \text{fpop}_{ij} + \beta_2 x_{ij} + \beta_3 y_{ij} + \beta_4 (x \times y)_{ij} + u_j + \epsilon_{ij}$$

where β_4 represents the interaction effect of x and y , capturing spatial properties.

```
Data: data_drop
Models:
fitlmer0: spf ~ 1 + (1 | year)
fitlmer1: spf ~ fpop + (1 | year)
fitlmer2: spf ~ fpop + x + (1 | year)
fitlmer3: spf ~ fpop + x + y + (1 | year)
fitlmer4: spf ~ fpop + x + y + x * y + (1 | year)

      npar    AIC    BIC logLik deviance   Chisq Df Pr(>Chisq)
fitlmer0     3 2778.1 2791.4 -1386.0   2772.1
fitlmer1     4 2768.2 2786.1 -1380.1   2760.2 11.8419  1 0.0005791 ***
fitlmer2     5 2759.6 2781.8 -1374.8   2749.6 10.6806  1 0.0010827 **
fitlmer3     6 2761.5 2788.3 -1374.8   2749.5  0.0358  1 0.8500033
fitlmer4     7 2763.2 2794.3 -1374.6   2749.2  0.3776  1 0.5388740
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Table 8: Comparison of Linear Mixed Effects Models with Different Fixed Effects and Random Intercept for each grouping of **year**.

Model	Residual Std.Dev.
fitlmer0	2.0710
fitlmer1	2.0726
fitlmer2	2.0553
fitlmer3	2.0552
fitlmer4	2.0549

Table 9: Standard Deviations of Residuals for Models fitlmer0 to fitlmer4

From Table 8, it is evident that the progression from fitlmer0 to fitlmer4 follows an incremental approach, starting with the simplest model and gradually incorporating more complexity and nesting. The objective is to identify a model that is sufficiently complex to explain the variability of **spf**, while ensuring that added complexity is statistically improving our model. Adding the **x** variable to fitlmer1 results in a statistically significant improvement, as indicated by the likelihood ratio test with $\text{Pr}(>\text{Chisq}) = 0.0010827$.

However, subsequent additions, including the $\mathbf{y}$ variable and the interaction between $\mathbf{x}$ and $\mathbf{y}$, do not provide any further significant improvement to the model.

Furthermore, the BIC achieves a local minimum at `fitlmer2`, suggesting it strikes a balance between model complexity and goodness-of-fit. As shown in Table 9, the residual standard deviations for `fitlmer1` and `fitlmer2` are very close, at 2.0726 and 2.0553, respectively. Although adding the $\mathbf{y}$ variable slightly reduces the residual standard deviation, it does not yield a significant improvement in model fit. Based on these criteria, `fitlmer2` is considered the better-performing model in this sequence.

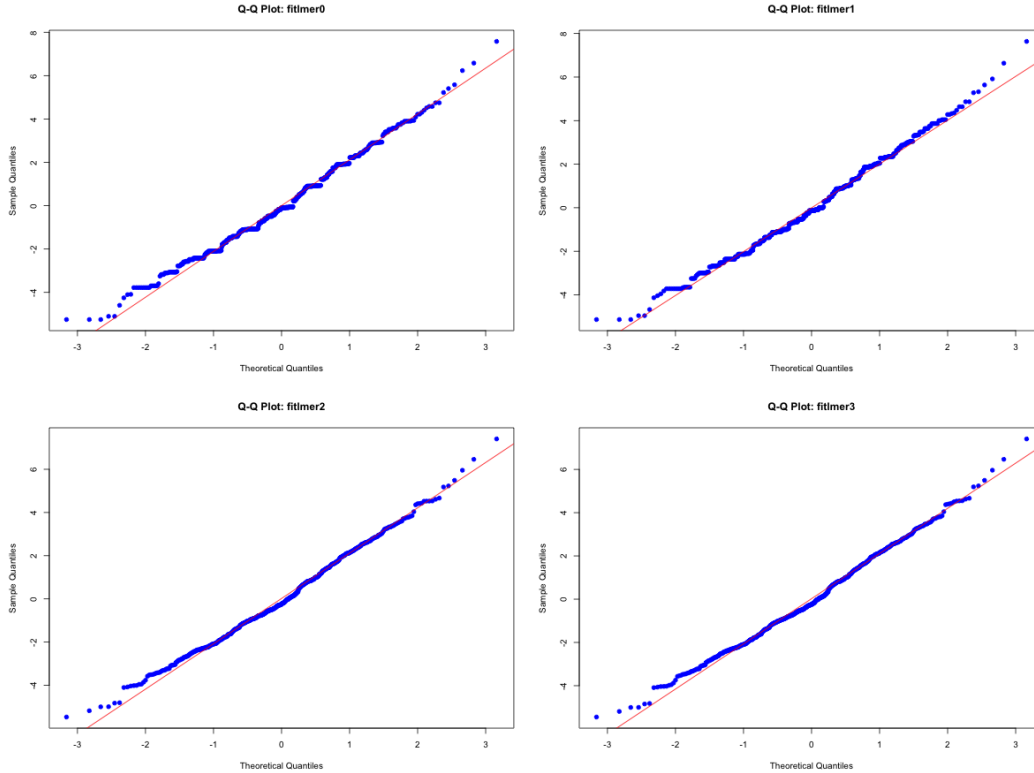


Figure 8: Q-Q plots of residuals for the linear mixed-effects models (`fitlmer0` to `fitlmer3`). These plots demonstrate improved adherence to the normality assumption compared to the linear models, particularly due to the incorporation of random effects for **year**, which accounts for grouping effects.

The Q-Q plots in Fig. 8 show the residuals of the linear mixed-effects models (`fitlmer0` to `fitlmer3`). These models incorporate random effects for **year**, addressing the grouping effects that violate the independence assumption in the linear models. Compared to the Q-Q plots for the linear models, the residuals here align more closely with the theoretical quantiles, indicating improved adherence to the normality assumption. Hence, the mixed-effects models better capture the variability of **spf** by accounting for the hierarchical structure. However, the slight deviation of the tail behavior is an incentive for us to go deeper with adding the random slope to our `lmer` model. Our goal is to extend it to a more generalized scenario.

4.2.2 Random Intercept and Random Slope

- **Model `fitlmer0_slope`: Null Model with Random Slope**

$$\text{spf}_{ij} = \beta_0 + u_{0j} + u_{1j}\text{fpop}_{ij} + \epsilon_{ij}$$

where $u_{0j} \sim \mathcal{N}(0, \Psi_{u_0}^2)$, $u_{1j} \sim \mathcal{N}(0, \Psi_{u_1}^2)$, and $\epsilon_{ij} \sim \mathcal{N}(0, \sigma^2)$.

- **Model `fitlmer1_slope`: Adding Fixed Effect for `fpop`**

$$\text{spf}_{ij} = \beta_0 + \beta_1\text{fpop}_{ij} + u_{0j} + u_{1j}\text{fpop}_{ij} + \epsilon_{ij}$$

- **Model fitlmer2_slope: Adding Fixed Effect for x**

$$\text{spf}_{ij} = \beta_0 + \beta_1 \text{fpop}_{ij} + \beta_2 x_{ij} + u_{0j} + u_{1j} \text{fpop}_{ij} + \epsilon_{ij}$$

- **Model fitlmer3_slope: Adding Fixed Effect for y**

$$\text{spf}_{ij} = \beta_0 + \beta_1 \text{fpop}_{ij} + \beta_2 x_{ij} + \beta_3 y_{ij} + u_{0j} + u_{1j} \text{fpop}_{ij} + \epsilon_{ij}$$

- **Model fitlmer4_slope: Adding Interaction between x and y**

$$\text{spf}_{ij} = \beta_0 + \beta_1 \text{fpop}_{ij} + \beta_2 x_{ij} + \beta_3 y_{ij} + \beta_4 (x_{ij} \cdot y_{ij}) + u_{0j} + u_{1j} \text{fpop}_{ij} + \epsilon_{ij}$$

```
data_drop
Models:
fitlmer0_slope: spf ~ 1 + (fpop | year)
fitlmer1_slope: spf ~ fpop + (fpop | year)
fitlmer2_slope: spf ~ fpop + x + (fpop | year)
fitlmer3_slope: spf ~ fpop + x + y + (fpop | year)
fitlmer4_slope: spf ~ fpop + x + y + x * y + (fpop | year)
      npar    AIC    BIC logLik deviance Chisq Df Pr(>Chisq)
fitlmer0_slope      5 2781.2 2803.5 -1385.6   2771.2
fitlmer1_slope      6 2773.2 2799.9 -1380.6   2761.2 9.9777  1  0.001584 **
fitlmer2_slope      7 2768.6 2799.8 -1377.3   2754.6 6.6069  1  0.010158 *
fitlmer3_slope      8 2765.4 2801.0 -1374.7   2749.4 5.2411  1  0.022060 *
fitlmer4_slope      9 2768.4 2808.5 -1375.2   2750.4 0.0000  1  1.000000
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Table 10: Comparison of Linear Mixed Effects Models with Different Fixed Effects, Random Intercept and Random Slope for each grouping of **year**.

Model	Residual Std.Dev.
fitlmer0_slope	2.0733
fitlmer1_slope	2.0762
fitlmer2_slope	2.0474
fitlmer3_slope	2.0520
fitlmer4_slope	2.0586

Table 11: Residual standard deviations for random slope models (fitlmer0_slope to fitlmer4_slope).

After incorporating the random slope into our model, we obtain the results shown in Table 11, which indicate a slight improvement in the residual standard deviation. However, in Table 10, we observe that the best model is fitlmer2_slope with BIC = 2799.8. When compared to the BIC of fitlmer2 without the random slope, which is 2781.8, the performance does not improve despite the increased complexity. Therefore, we conclude that under the lmer setting, the random slope does not contribute to explaining the variability of **spf**. Additionally, the Q-Q plots in Fig. 9 do not show significant differences compared to the lmer model with only the random intercept. This slight deviation in the tail behavior motivates further exploration into generalizing the model to a more flexible generalized mixed-effects model.

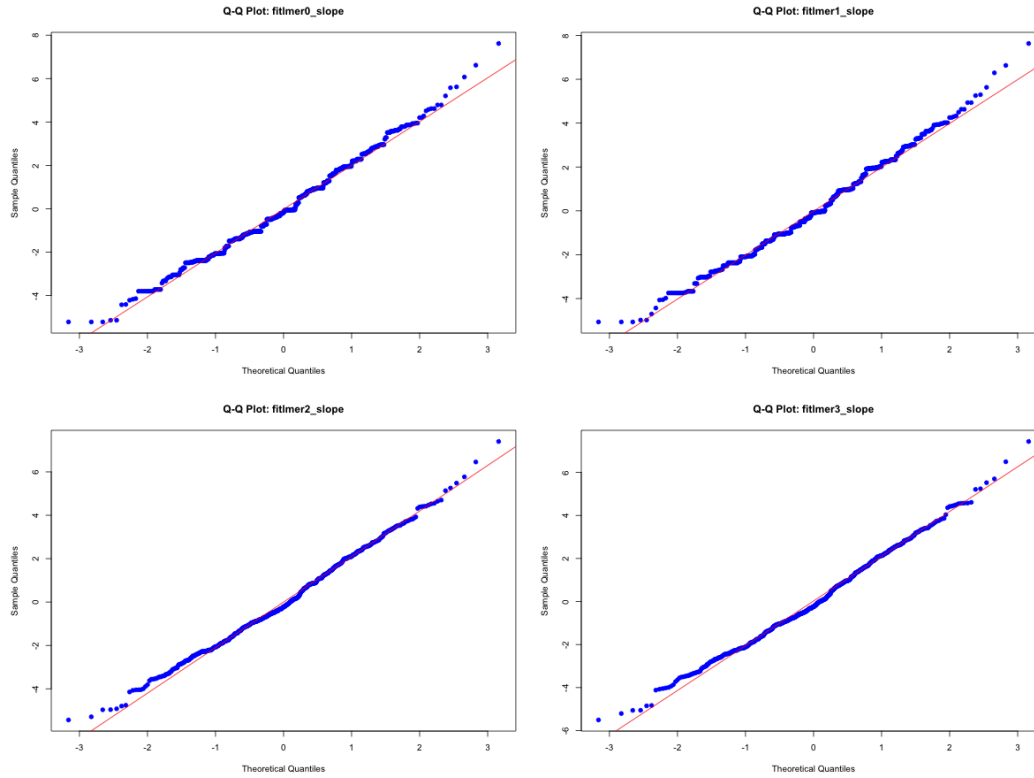


Figure 9: Q-Q plots of residuals for the linear mixed-effects models(random intercept and random slope) (fitlmer0 to fitlmer3).

4.3 Generalized Mixed effect model

The extension from `lmer` to `glmer` is motivated by the nature of the response variable, **spf**, which represents the count data and discrete. Linear mixed-effects models (`lmer`) assume a normal distribution for the residuals, which is strict and may not be suitable for count data as such data are often skewed and have non-negative integer values. Even though our residual plot in `lmer` does not have a huge deviation, making the model hunting process clear would be great for deciding our optimal model. In the paper, we adopt the Poisson distribution, which is normally used to count data. This transition ensures that the model assumptions align with the characteristics of the data, leading to more reliable and interpretable results. However, some point worth mentioning is that the complicated model does not guarantee better capturing the variability of **y** variable. Which should be diagnosed further.

fitglmer0_slope: spf ~ 1 + (fpop year)									
fitglmer1_slope: spf ~ fpop + (fpop year)									
fitglmer2_slope: spf ~ fpop + x + (fpop year)									
fitglmer3_slope: spf ~ fpop + x + y + (fpop year)									
fitglmer4_slope: spf ~ fpop + x + y + x * y + (fpop year)									
	npar	AIC	BIC	logLik	deviance	Chisq	Df	Pr(>Chisq)	
fitglmer0_slope	4	2751.2	2769.1	-1371.6	2743.2				
fitglmer1_slope	5	2742.8	2765.1	-1366.4	2732.8	10.4091	1	0.0012539	**
fitglmer2_slope	6	2731.9	2758.6	-1359.9	2719.9	12.9599	1	0.0003182	***
fitglmer3_slope	7	2735.1	2766.3	-1360.5	2721.1	0.0000	1	1.0000000	
fitglmer4_slope	8	2735.4	2771.1	-1359.7	2719.4	1.6554	1	0.1982297	

Table 12: Comparison of Generalized Linear Mixed Effects Models with Different Fixed Effects, Random Intercept, and Random Slope for each grouping of **year** using a Poisson distribution.

Model	Residual_SD
fitglmer0_slope	1.133463
fitglmer1_slope	1.135064
fitglmer2_slope	1.124027
fitglmer3_slope	1.124355
fitglmer4_slope	1.123698

Table 13: Residual standard deviations for generalized linear mixed models with Poisson distribution using random intercept and slope for **year**.

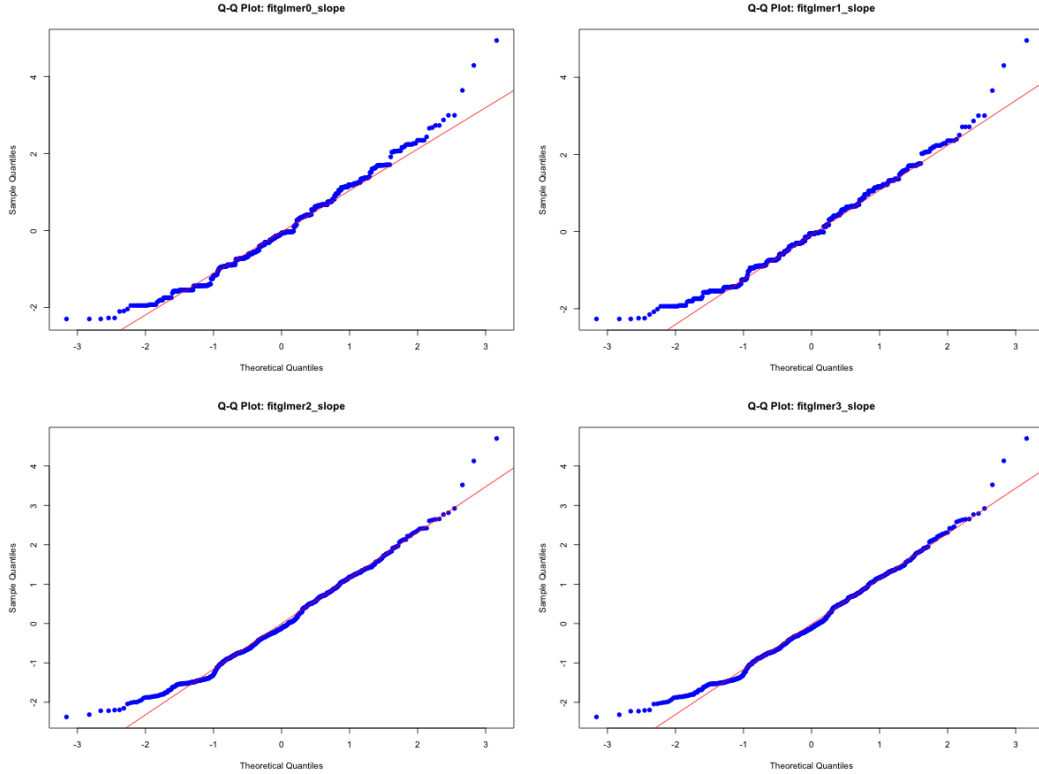


Figure 10: Q-Q plots of residuals for GLMMs with random slope under the Poisson model.

Based on Table 12, we observe a further decrease in BIC, which suggests improvement. Additionally, the reduction in residual standard deviation, as shown in Table 13, indicates that the glmer model performs better in terms of model fit. However, as noted in [Hoff, NA], specifically in s13ModelChecksNP, the normality assumption is critical for valid inference. From Fig. 10, the normality assumption appears to deviate more than in the lmer model. Therefore, we question the validity of the estimation and inference methods based on the glmer model. As a result, we select fitlmer2 as our final model.

4.4 Final Model For Interpretation

- Final Model:

- Model fitlmer2:

$$\text{spf}_{ij} = \beta_0 + \beta_1 \cdot \text{fpop}_{ij} + \beta_2 \cdot x_{ij} + u_j + \epsilon_{ij}$$

where:

- * β_0 : Overall intercept.
- * β_1 : Fixed effect of fpop_{ij} .
- * β_2 : Fixed effect of x_{ij} .

- * $u_j \sim N(0, \Psi_u^2)$: Random intercept for group (year) j .
- * $\epsilon_{ij} \sim N(0, \Psi_\epsilon^2)$: Residual error for observation i in group j .

- Assumption Check:

- Normality: The Q-Q plot in Fig. 11 shows that the residuals of the final model (fitlmer2) largely follow the theoretical normal line, though some deviations are observed at the tails, particularly on the upper end. This indicates that the residuals approximately satisfy the normality assumption, which is crucial for valid inference in linear mixed-effects models. The histogram of residuals complements this finding, with the residuals displaying a roughly symmetric distribution centered around zero. While minor deviations from normality exist, they are not severe enough to question the reliability of the final model for interpretation purposes.

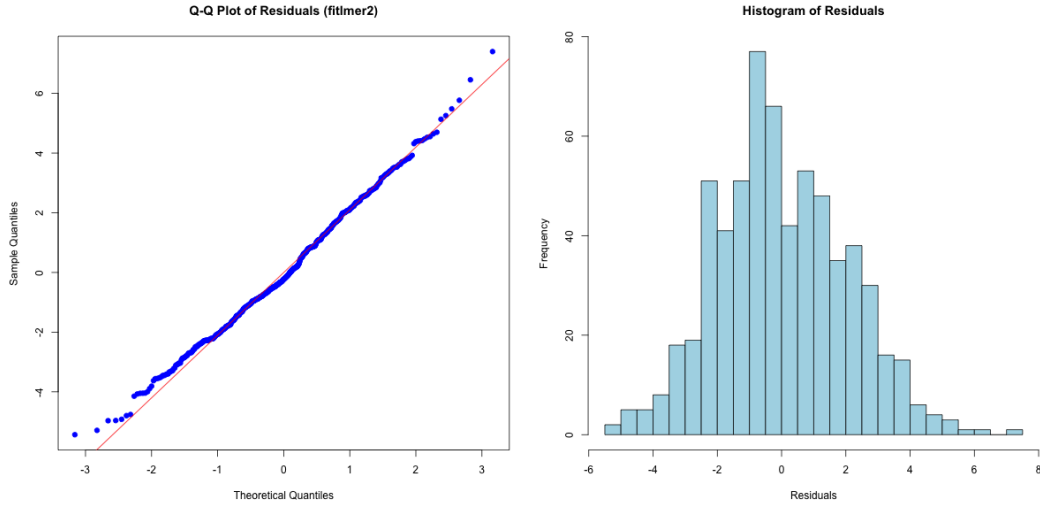


Figure 11: Q-Q plot (left) and histogram (right) of residuals for the final model (fitlmer2). The Q-Q plot assesses the normality of residuals by comparing the observed residuals to a theoretical normal distribution, while the histogram provides a visual representation of the residual distribution.

- Heteroscedasticity: The boxplot in Fig. 12 illustrates the distribution of residuals grouped by **year**. While the spread of residuals appears to vary slightly between years, there is no consistent pattern indicating severe heteroscedasticity. The variability within some years is larger than others, but this fluctuation is expected given the hierarchical nature of the data. Overall, the residuals seem reasonably balanced across the grouping factor, suggesting that the assumption of homoscedasticity is not strongly violated.

5 Model and data analysis interpretation

5.1 Fixed Effects Parameters

From the results of the final model, we interpret the fixed effects as follows:

- **Intercept** ($\beta_0 = 6.699$): The estimated average number of fledged sparrows (**spf**) is approximately 6.699 when both **fpop** (number of females in the population) and **x** (x-coordinate of the nest) are at zero. This represents the baseline reproductive success in the absence of other effects.
- β_1 (**fpop effect**, -0.053): For each additional female in the population, the number of offspring fledged decreases by approximately 0.053 on average. This negative coefficient suggests that competition among females has a detrimental effect on reproductive success.

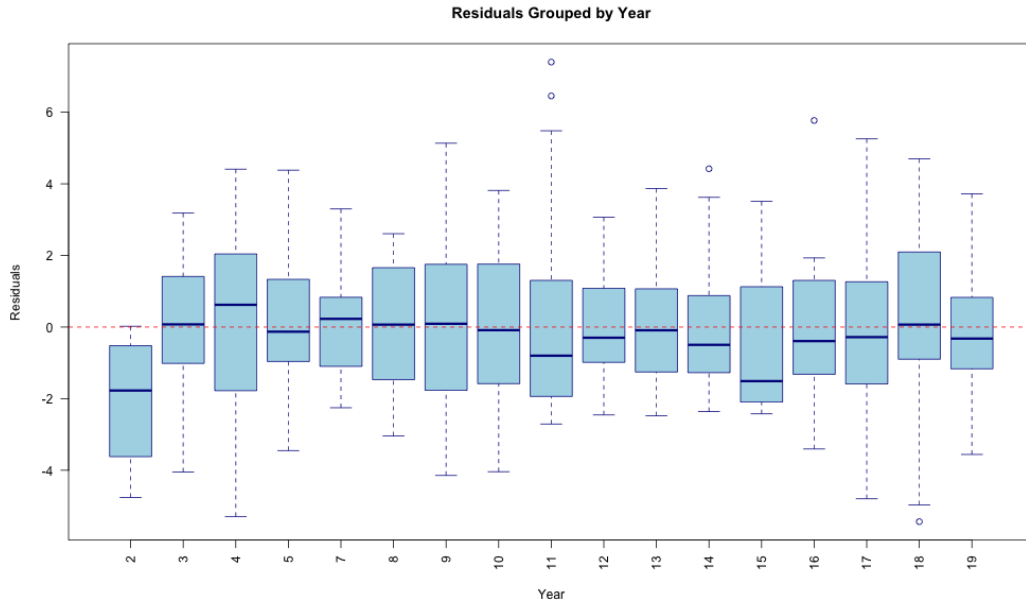


Figure 12: Boxplot of residuals grouped by **year** for the final model (fitlmer2). Each box represents the distribution of residuals within a specific year, showing the variation and spread of residuals across groups.

- β_2 (**x effect**, -0.034): For each unit increase in the x-coordinate of the nesting site, the number of offspring fledged decreases by approximately 0.034 on average. This suggests that certain nesting locations, characterized by their x-coordinate, may be less favorable for reproductive success.

5.2 Confidence Intervals for Fixed Effects

Given the p-values for all fixed effects (< 0.01), we are confident that these effects are statistically significant. The confidence intervals for these coefficients would exclude zero, reinforcing their impact on the response variable (**spf**).

5.3 Random Effects

The random intercept variance for **year** (0.584) indicates variability in reproductive success across different years. This highlights how yearly environmental or ecological conditions may influence the number of fledglings.

5.4 Implications and Findings

- **Competition Among Females:** The negative effect of **fpop** confirms that reproductive success diminishes as the population of females increases, likely due to resource competition or territorial disputes.
- **Nesting Site Influence:** The effect of **x** suggests that spatial factors play a role in fledging success. Certain areas of the island may provide suboptimal conditions, such as reduced food availability or higher predation risk.
- **Yearly Variability:** The significant random effect of **year** emphasizes the importance of considering environmental factors that vary over time, such as weather or food abundance, when studying reproductive outcomes.

Till now, we can say the final model effectively captures the key determinants of reproductive success among song sparrows, illustrating how competition, spatial factors, and temporal variability shape the number of offspring fledged.

6 Conclusion

This sparrow dataset analysis shown some key insights into factors affecting reproductive success. Most of the scenario match with our human’s intuition. For instance, our model identified competition among females and spatial factors as significant determinants of the number of fledged sparrows. However, only 1 spatial variable is statistically significant.

The negative slope or effect of the number of females in the population (**fpop**) is a sign saying that higher competition among females for resources or nesting sites reduces reproductive success. This should be true as the competition would make the nest lack resources or have less nutrition(bugs) could be hunted within the area. Similarly, the spatial effect represented by the x-coordinate (**x**) implies that certain nesting areas are less favorable, or, say, detrimental, potentially due to high predation risks, wind, water, or affection by other factors.

6.1 Hypothesis Tests and Model Selection

The hypothesis tests for fixed effects in the final model (**fitlmer2**) confirm that both **fpop** and **x** significantly influence reproductive outcomes. Model selection, based on BIC and AIC criteria, demonstrated that **fitlmer2**, which includes **fpop** and **x**, is the most suitable among the tested models. The random intercept for **year** highlights inter-annual variability, which could be extrapolated that in certain years, it might be due to the ecological, environmental, or any metrological factors that make the reproductive pattern particularly that way in that specific year. Hence, taking it as the grouping factor should be a reasonable choice.

6.2 Limitations

Despite its strengths, the analysis has limitations. The normality assumption for residuals showed very slight deviations in both sides of the tails, which could affect inference accuracy. Also, the spatial data does not clearly explain what **x** and **y** stands for, in terms of spatial meaning, this should be a critical information. Another limitation is the inability to fully address temporal trends due to potential confounding with environmental variables not included in the model.

6.3 Future Directions

Aside from the more advanced statistical learning methods, I will say the experimental design could take other factors into account. For instance, the environmental factors such as temperature for that year or any invasive species, the longevity of the observation, and frequency, will make the research more accurate. It is likely for the offspring to die immediately for many reasons when they just give birth; hence, when the observer comes, the number tagged would not be accurate. Taking this into account should be important.

In summary, in our main findings, the spatial and competition from other females matter for the number of offspring. Some possible improvement including checking for the first year of observation; the possible explanation of incorrect chrononogical order. Furthermore, we could also do the data-generating process to combine **x** and **y**, which may be possible to bring a different story to the variation of the number of offspring.

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This report uses Grammarly, ChatGPT to polish grammar.

7 Replication code: R Markdown